



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Complete Spaces

Completeness of $\mathbb{R}^n$, ℓ^p , ℓ^∞ and $C[0, 1]$

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CONDUCTED BY

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Completeness of a Metric Space

Complete Metric Space

A metric space (X, d) is called **complete** if every Cauchy sequence in X converges to a point in X .

Completeness of $\mathbb{R}^n$

💡 $\mathbb{R}$ is a Complete Metric Space

The set of real numbers $\mathbb{R}$ with the usual metric $d(x, y) = |x - y|$ is a complete metric space. 💡

💡 $\mathbb{Q}$ is not a Complete Metric Space

The set of rational numbers $\mathbb{Q}$ with the usual metric $d(x, y) = |x - y|$ is not a complete metric space. 💡

💡 $\mathbb{R}^n$ is a Complete Metric Space

The set of n -tuples of real numbers $\mathbb{R}^n$ with the usual metric



$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

is a complete metric space.

Completeness of ℓ^p Spaces

💡 ℓ^p is a Complete Metric Space

The set of sequences ℓ^p with the metric

$$d(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^{\infty} |x_i - y_i|^p \right)^{1/p} \quad \mathbf{x}, \mathbf{y} \in \ell^p$$

is a complete metric space.



💡 ℓ^∞ is a Complete Metric Space

The set of sequences ℓ^∞ with the metric

$$d(\mathbf{x}, \mathbf{y}) = \sup_{i \in \mathbb{N}} |x_i - y_i| \quad \mathbf{x}, \mathbf{y} \in \ell^\infty$$

is a complete metric space.



Completeness of Function Spaces

💡 Cauchy criterion for uniform convergence

Let $\{f_n\}$ be a sequence of functions defined on a set A and taking values in a metric space (Y, d) . The sequence $\{f_n\}$ converges uniformly to a function f on A if and only if for every $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ such that for all $m, n \geq N$ and for all $x \in A$, the following inequality holds:

$$d(f_n(x), f_m(x)) < \varepsilon.$$

$C[0, 1]$ under sup metric is a Complete Metric Space

Let $C[0, 1]$ be the set of all continuous real-valued functions defined on the interval $[0, 1]$. Define a metric d on $C[0, 1]$ by

$$d(f, g) = \sup_{x \in [0, 1]} |f(x) - g(x)|$$

for all $f, g \in C[0, 1]$. Show that $(C[0, 1], d)$ is a complete metric space.

❓ $C[0, 1]$ under integral metric is not Complete

Let $C[0, 1]$ be the set of all continuous real-valued functions defined on the interval $[0, 1]$. Define a metric d on $C[0, 1]$ by

$$d(f, g) = \int_0^1 |f(x) - g(x)| dx$$

for all $f, g \in C[0, 1]$. Show that $(C[0, 1], d)$ is not a complete metric space.

Thank You!

We'd love your questions and feedback.

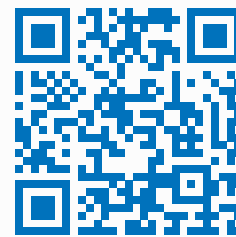
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(Lectures, walkthroughs, and course updates)



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References

- [1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.
- [2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.